Effective process management through performance measurement

Part III – an integrated model of total quality-based performance measurement

David Sinclair and Mohamed Zairi
The European Centre for TQM, Bradford University, Bradford, UK

Introduction

The purpose of this article is to introduce a model for total quality-based performance measurement systems, based on case study research outlined in parts I and II[1,2]. The model is in turn validated by the results from a postal survey of the performance measurement systems in 115 companies.

It is important to note that in most organizations there is no separate "performance measurement system". The model therefore integrates measurement within the overall management process. The model consists of five levels: strategy development and goal deployment; process management and measurement; performance appraisal and management; break-point performance assessment; and reward and recognition systems.

Definition of terms

Before examining the model in detail, it is useful to formally define the terms used in the model. The following terms are generally applicable to the model:

- *Performance measurement* is "the process of determining how successful organizations or individuals have been in attaining their objectives"[3]. Total quality-based performance measurement can be defined as the measurement of non-financial performance at all levels within the organization (including individuals, teams, processes, departments and the organization as a whole), with a view to the continuous improvement of performance against organizational objectives.
- *Performance measures* are "the numerical or quantitative indicators that show how well each objective is being met"[4].
- *A performance measurement system* is "a systematic way of evaluating the inputs, outputs, transformation and productivity in a manufacturing or non-manufacturing operation"[5]. A total quality-based performance measurement system can be defined as a system which integrates the measurement of non-financial performance at all levels within the organization with a view to the continuous improvement of performance against organizational objectives[6].

Business Process Re-engineering
& Management Journal,
Vol. 1 No. 3, 1995, pp. 50-65.

- *A target* can be defined as the predetermined desired level of performance against each measure.

The following terms are specific to each level of the model:

(1) Strategy development and goal deployment:
   - Critical success factors (CSF) are "the limited number of areas in which results, if they are satisfactory, will ensure successful competitive performance for the organization"[7].
   - Key performance indicators (KPI) are the actual measures used to quantitatively assess performance against the critical success factors. There should be at least one KPI for each critical success factor.

(2) Process management and measurement. It is suggested that process performance measures should include measures of inputs, process and outputs. Outputs from a supplier form the inputs to the next customer. It is therefore only necessary to define process and output measures:
   - Process measures "monitor the activities of a process and motivate people within a process"[8].
   - Output measures "report the results of a process, often to management, and are used to control resources"[8].

(3) Performance appraisal and management. Performance appraisal and management includes the two linked elements of performance management and performance appraisal:
   - performance management is "a systematic, data-oriented approach to managing people at work that relies on positive reinforcement as the major way to maximize performance"[9]. Performance management refers to the management of individuals and teams on a frequent and ongoing basis.
   - Performance appraisal is "the process by which organizations establish measures and evaluate individual employees' behaviour and accomplishments for a finite period of time"[10].

(4) Break-point performance assessment can be defined as the measurement of any performance criteria which is intended to identify significant gaps in current performance, and thereby motivate activities to improve performance so as to reduce or eliminate the gap[6]. Break-point assessment includes both internal (within the organization) and external (involving sources outside the organization) measurement techniques.

(5) Reward and recognition are the financial and non-financial consequences given as a result of superior performance[6]:
   - Rewards are the *financial* consequences given as the result of measurably superior performance.
   - Recognition includes all *non-financial* consequences given as the result of measurably superior performance.

Strategy development and goal deployment

The first "level" of the performance measurement system model is the development of organizational strategy, and the consequent deployment of goals throughout the organization. Steps in the strategy development and goal deployment process are[6]:

(1) Develop a public mission statement based on recognizing the needs of all organizational stakeholders (customers, employees, shareholders and society). This includes mission statement, vision statement, quality policy, and corporate values.

(2) Based on the mission statement, identify those factors critical to the success of the organization achieving its stated mission (i.e. identify critical success factors or CSFs). CSFs should represent all stakeholder groups (customers, employees, shareholders and society).

(3) Define performance measures for each CSF – i.e. key performance indicators (KPI). There may be one or several KPIs for each CSF. In some cases, organizations may define CSFs and KPIs separately, such that CSFs are set to be constant, or they may be identical to KPIs, and therefore be expected to change over time. Definition of KPIs should include:

- title of KPI;
- data used in calculation of KPI;
- method of calculation of KPI;
- sources of data used in calculation;
- proposed measurement frequency;
- responsibility for the measurement process.

(4) Set targets for each KPI. If KPIs are new, targets should be based on customer requirements, competitor performance or known organizational criteria. If no such data exists, a target should be set based on "best guess" criteria. If the latter is used, update the target as soon as enough data is collected to be able to do so.

(5) Assign responsibility at the organizational level for achievement of desired performance against KPI targets. Responsibility should rest with directors and very senior managers.

(6) Develop plans to achieve the target performance. This includes both action plans for one year, and longer-term strategic plans.

(7) Deploy mission, CSFs, KPIs, targets, responsibility and plans to macro processes (core business processes). This includes the communication of goals, objectives' plans, and the assignment of responsibility with the appropriate individuals.

(8) Manage organizational processes (see Level 2 of the model).

(9) Measure performance against organizational KPIs, and compare to target performance.

(10) Based on this comparison, identify areas with high leverage for improvement, and update action plans.

(11) Communicate performance and proposed actions throughout the organization.

(12) At the end of one year compare organizational capability to target against all KPIs, and begin again at Step 2 above.

(13) Reward and recognize superior organizational performance.

Strategy development and goal deployment is the responsibility of senior management within the organization, although there should be as much input to the process as possible by both experts in the area, and employees generally, in order to achieve "buy-in" to the process. Strategy development and goal deployment (Level 1 of the performance measurement system model) is shown diagramatically in Figure 1.

It should be noted that financial performance for external reporting purposes is seen as a result of performance against non-financial KPIs, as suggested by Walsh[11] and Kaplan and Norton[12]. The only aspect of financial performance which is cascaded throughout the organization is the budgetary process, which acts as a constraint rather than a performance improvement measure, as suggested by Euske *et al.*[13].

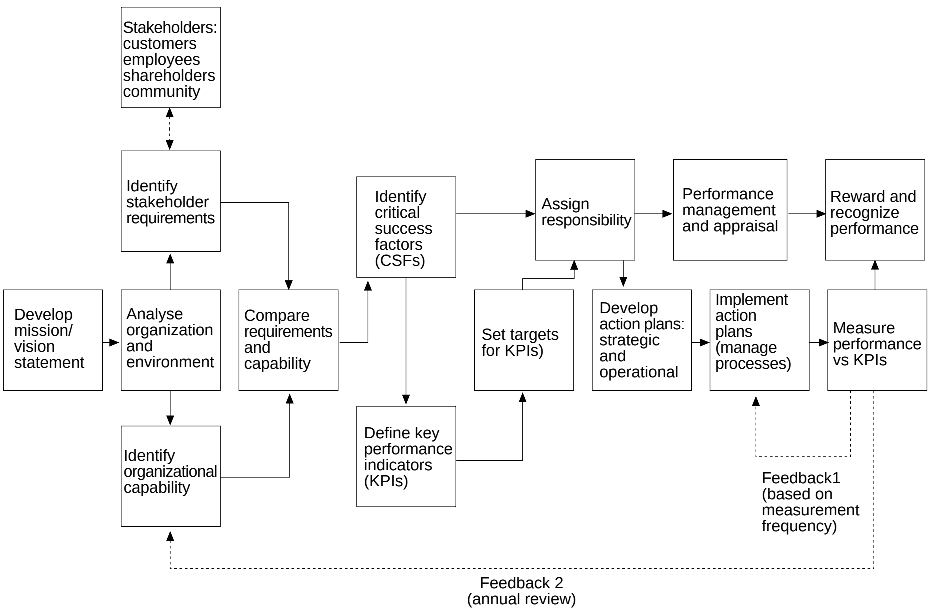

Figure 1. Performance measurement system model Level 1: strategy development and goal deployment

The difference between the KPIs at the process and organizational level is due to the deployment of organizational KPIs. Not only measures, but also CSFs, plans and responsibility are cascaded to the process level. As such the process is nearer to the quality policy deployment (QPD) approach, suggested by Zairi[14] and Oakland[15].

Process management and measurement

The second level of the performance measurement system model is process management and measurement. Steps in process management and measurement are[6]:

(1) If not already completed, identify and map processes. This information should include identification of:
- process customers and suppliers (internal and external);
- customer requirements (internal and external);
- core and non-core activities;
- measurement points and feedback loops.

(2) Translate organizational goals and action plans and customer requirements into process performance measures (input, in-process and output). This includes definition of measures, data collection procedures, and measurement frequency.

(3) Define appropriate performance targets (based on known process capability, competitor performance and customer requirements).

(4) Assign responsibility for achieving performance targets.

(5) Develop plans towards achievement of process performance targets.

(6) Deploy measures, targets, plans and responsibility to all sub-processes.

(7) Operate processes.

(8) Measure performance against process KPIs and compare with target performance.

(9) Use performance information to:
- implement continuous improvement activities;
- identify areas for improvement;
- update action plans;
- update performance targets;
- redesign processes (where appropriate);
- manage the performance of teams and individuals (performance management and appraisal) and suppliers;

- provide leading indicators and explain performance against organizational KPIs.

(10) At the end of each year compare process capability to customer requirements against all measures, and begin again at Step 2.

(11) Reward and recognize superior process (including sub-processes and teams) performance.

The above process should be managed by the process owner, with inputs wherever possible from the owners of sub-processes. Process management and measurement (Level 2 of the performance measurement system model) is shown diagrammatically in Figure 2.

The process outlined should be used whether an organization is organized and managed on a process or functional (departmental) basis. If the organization is functionally organized, the key task is to identify the customer-supplier relationships between functions, and for functions to see themselves as part of a customer-supplier chain[15].

Performance measures

While some organizations measure performance along the same dimensions as at the organizational level, using some kind of balanced scorecard approach, other organizations monitor performance across different dimensions according to the process. At all organizations, however, measurements can be identified as input (supplier), in-process, and output (or results).

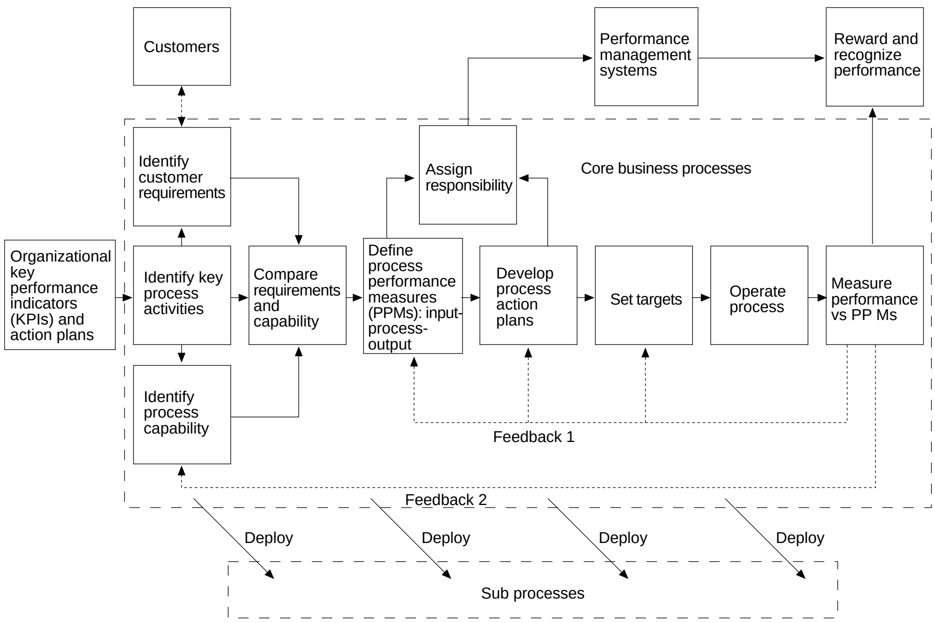

Figure 2. Performance measurement system model Level 2: process management and measurement

Summary

Process management and measurement has been well documented in the literature on TQM. However, at the organizations examined in this study there were great differences between the levels of management of processes, as opposed to the management of functions[6]. Cross-functional performance measurement is a vital component of the removal of "functional silos", and the consequent potential for sub-optimization and failure to take account of customer requirements[16].

Performance appraisal and management

The third level of the performance measurement system model is the management of individuals. Steps in performance appraisal and management are[6]:

(1) If not already completed, identify and document job descriptions based on process requirements and personal characteristics. This information should include identification of:
- activities to be undertaken in performing the job;
- requirements of the individual with respect to the identified activities, in terms of experience, skills and training;
- requirements for development of the individual, in terms of personal training and development.

(2) Translate process goals and action plans and personal training and development requirements into personal performance measures.

(3) Define appropriate performance targets based on known capability and desired characteristics (or desired characteristics alone if there is no prior knowledge of capability).

(4) Develop plans towards achievement of personal performance targets.

(5) Document 1 to 4 using appropriate forms. Documentation should include space for the results of performance appraisal.

(6) Manage performance. This includes:
- planning tasks on a daily/weekly basis;
- managing performance of the tasks;
- monitoring performance against task objectives using both quantitative (process) and qualitative information on a daily and/or weekly basis;
- giving feedback to individuals of their performance in carrying out tasks;
- giving recognition to individuals for superior performance.

(7) Formally appraise performance against the range of measures developed, and compare with target performance.

(8) Use comparison with target to:
- identify areas for improvement;
- update action plans;
- update performance targets;
- redesign jobs (where appropriate) – this impacts Step 1 of the process.

(9) Update documentation.

(10) After a suitable period (ideally more than once a year) compare capability to job requirements, and begin again at Step 2.

(11) Reward and recognize superior performance.

The above activities should be undertaken by the individual whose performance is being managed, together with their immediate superior. Performance appraisal and management is shown diagramatically in Figure 3.

The major difference between TQ and non-TQ approaches in the management of individuals appears to lie in the reward of effort as well as achievement and the consequently different measures used[17], the use of performance management in TQM[7], the use of information in continuous improvement and to reward and recognize performance in TQ-organizations[15], and the attempt to manage teams in TQ-organizations[18].

The organizations examined in this study attempted to measure a combination of process/task performance (effort and achievement) and personal development[6]. The frequency of formal performance appraisal is defined by

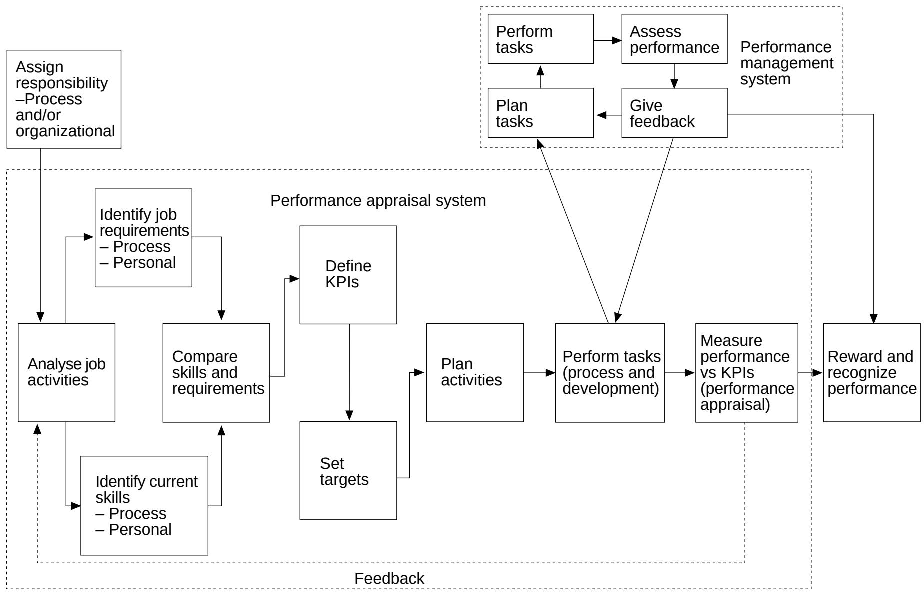

Figure 3. Performance measurement system model Level 3: performance appraisal and management

the frequency of the appraisal process (generally with a minimum frequency of six months). Most organizations recognized that this period was probably too long, but none suggested that they would increase the frequency. Between formal performance reviews, organizations rely on performance management techniques to manage individuals.

"Break-point" performance assessment

The fourth level of the performance measurement system model is break-point assessment. Steps in break-point performance assessment are as follows[6]:

(1) Identify the need for assessment. Identification of the need for assessment will come from:
- poor performance at the organizational or process levels against KPIs;
- identified superior performance of competitors;
- customer inputs;
- the desire to direct improvement efforts better;
- the desire to concentrate attention on the need for performance improvement.

(2) Identify mode and technique of assessment. This involves determining whether the assessment should be carried out internally within the organization, or externally, and the type of assessment that should be carried out. Some techniques are purely internal (e.g. self-assessment), while others (e.g. external benchmarking) involve obtaining information from sources external to the organization. The choice depends on:
- How the need for assessment was identified (see 1).
- The aim of the assessment. If the aim is to improve performance relative to competitors, external benchmarking may be a better option than internally measuring the cost of quality.
- The relative costs and expected benefits of each technique.

(3) Carry out the assessment.

(4) Feed results into the planning process at the organizational or process level.

(5) Determine whether to repeat the exercise. If it is decided to repeat the exercise, the following points should be considered:
- frequency of assessment;
- at what levels to carry out future assessment (e.g. organization-wide or process-by-process);
- decide whether the assessment technique should be incorporated into regular performance measurement processes, and if so how this will be managed.

The major differences between break-point assessment and other measures of performance are:

- Break-point assessment exercises are generally new to the organization, or are carried out less frequently than other measurement exercises.
- Break-point assessment often requires the use of a level of resources greater than that normally associated with performance measurement.
- Break-point assessment techniques give a broader view of performance than most individual measures.

Break-point performance assessment (Level 4 in the performance measurement system model) is shown diagramatically in Figure 4.

Techniques identified for break-point assessment at the organizations examined in this study include; quality costing, self-assessment (against Baldrige or European Quality Award criteria), benchmarking (internal or external), customer satisfaction surveys, quality function deployment (QFD) and activity-based costing (ABC)[6].

The use of the techniques as outlined differs from that described by most authors in two respects:

(1) The integration of break-point assessment techniques into the overall performance management process.

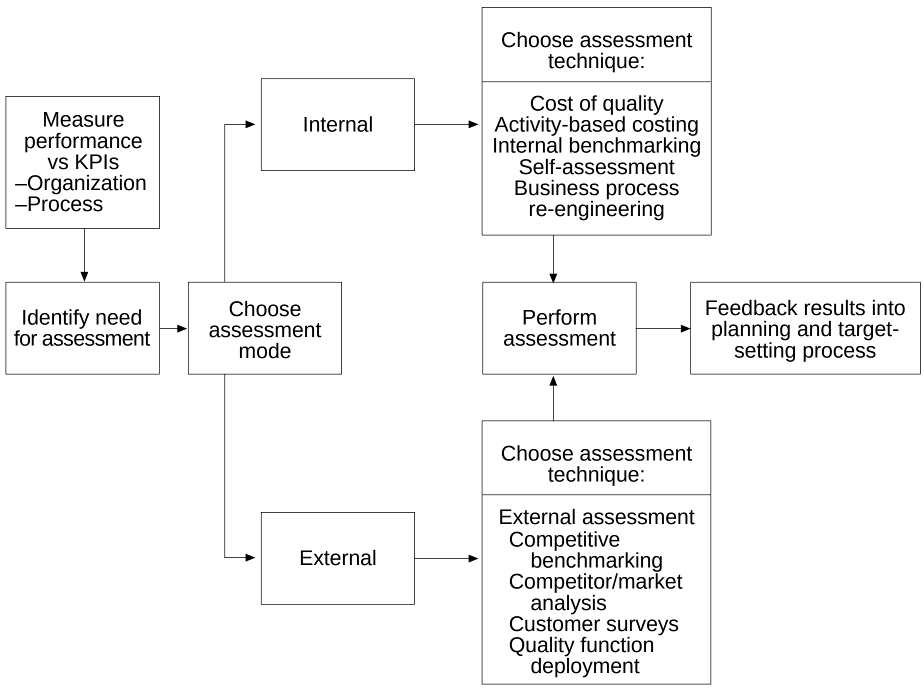

Figure 4. Performance measurement system model Level 4: break-point performance assessment

(2) The recognition that such techniques can move from being break-point techniques to being simply another organizational or process performance measure.

Reward and recognition systems

Reward and recognition form an output of performance measurement at the organisational, process (including suppliers), and performance appraisal and management (including performance-related pay and performance management) levels[6]. Reward and recognition appears to be often "bolted-on" to the management process. Reward and recognition is included in the performance measurement system model, since it would appear that all the organizations examined in this study have attempted to innovate in this area[6].

Summary

The overall performance measurement system model can be shown as a series of complementary PDCA cycles, as shown in Figure 5. It should be remembered that each cycle operates at a different frequency, and within each cycle there will be individual cycles for each measure. The frequency of the cycle is dependant on organizational level, process cycle time and criticality of the measurement.

The budgetary control process is separated from the outlined performance measurement system. The information in the performance measurement system is used in performance improvement, whereas budgetary control acts as a constraint within which performance is managed. This finding was also suggested by Euske *et al.*[13].

Validation of the model

The model developed on the basis of the analysis of the case study data needed to be validated for its practical applicability[6]. The aims of the survey were to:

- confirm that the model is in use at organizations which have implemented TQM;
- compare performance measurement in TQ and non-TQ organizations.

Sample selection

Questionnaires were sent to a random sample of 500 organizations. It was not possible to ensure that the sample included a balance of TQ and non-TQ organizations, since this information is not documented. It should also be remembered that there is little agreement on which organizations can truly be defined as "total quality" even when they claim to have implemented TQM.

Analysis of responses

In total 115 usable responses were returned, giving a response rate of 23 per cent. Of these, 95 organizations (83 per cent) had implemented TQM, while 20 (17 per cent) had no TQM process. The division between manufacturing and services was almost even, with 58 manufacturers and 57 services responding. The division of TQ and non-TQ organizations by industry was almost even. Of

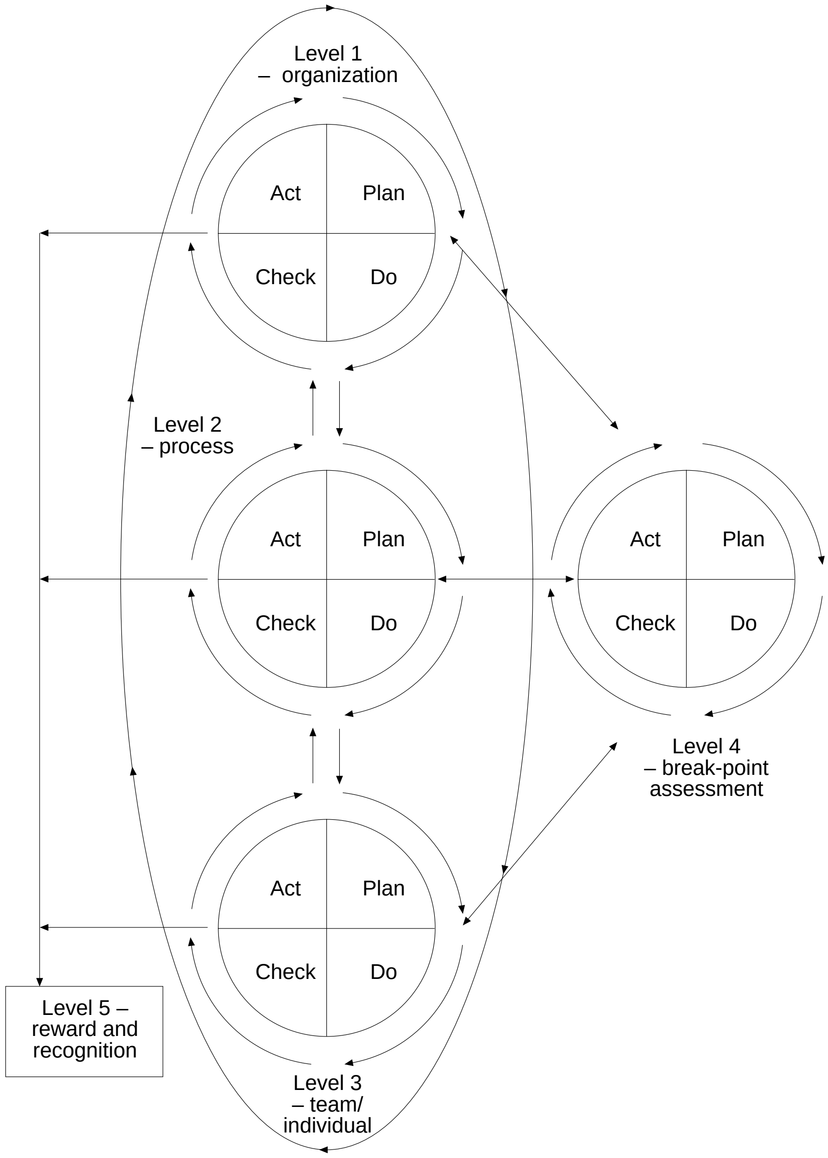

Figure 5.
Performance measurement system model

the 95 TQ organizations which responded, 49 (52 per cent) were manufacturers, and 46 (48 per cent) were services. Of the 20 non-TQ organizations, nine (45 per cent) were manufacturers, and 11 (55 per cent) were services.

Of the organizations which had implemented TQM, the average experience of use of TQM was 3.8 years. The majority of organizations (77 or 81 per cent) had less than five years' experience of TQM, although four organizations had at least ten years' experience of TQM. Manufacturers had on average one year more experience of TQM than services, with 4.2 years compared to 3.1 years.

In the following discussion of the survey, findings are generally descriptive, since the small number of non-TQ respondents (and difficulty defining true "TQ" organizations) precluded meaningful inferential statistical analysis[6].

Strategy development and goal deployment

In the first section of the questionnaire, respondents were asked to identify which of a number of strategic management practices had been implemented by their organization. The hypothesis was that TQ organizations were likely to have implemented a wider range of the formal strategic management practices identified at the organizations used to develop cases in this study[6].

Analysis of the data showed that, apart from the development of a mission/vision statement, TQ organizations in the sample are approximately twice as likely as non-TQ organizations to have implemented the strategic planning practices identified in the case study organizations. This includes development and communication of strategic objectives, the development of CSFs and KPIs, and the assignment of responsibility for strategic objectives.

Respondents were then asked to identify the purposes for which performance measurement is used in their organization. Purposes included measurement of organizational, process/functional, team and individual performance, goal alignment, continuous improvement, benchmarking and opportunity identification. Analysis of the data showed that TQ organizations use measurement significantly more widely than non-TQ organizations in the areas of overall company performance, process/functional performance, continuous improvement, benchmarking and opportunity identification. The use of measurement for team performance, individual performance and goal alignment are similar for TQ and non-TQ organizations.

Analysis of the data on performance measurement related to the strategy development and goal deployment level of the performance measurement system model identified clear differences between TQ and non-TQ organizations. TQ organizations are more likely to have formal strategic management processes based on the use of critical success factors and associated KPIs than non-TQ organizations. It should be remembered that such formal strategic control processes are associated with more effective performance measurement systems. The tentative conclusion which can be drawn from the above analysis is therefore that performance measurement systems in TQ organizations are likely to be more effective than in non-TQ organizations.

Process management and measurement

Respondents were asked to identify which aspects of processes have been identified, defined and documented in their organization. TQ organizations were found to be more likely to have mapped and documented all aspects of processes (core and sub-processes, process owners, suppliers and customers, measurement points, feedback loops, process performance measures, measurement frequency and performance targets) than non-TQ organizations. This is particularly the case in manufacturing organizations.

Respondents were asked to identify the individuals responsible for developing measures, setting performance targets, and actually measuring performance. Data analysis showed that in TQ organizations, responsibility for defining measures and measuring performance rests with process owners and their immediate superiors to a greater extent than for non-TQ organizations. In non-TQ organizations, responsibility rests to a greater extent with higher management. This suggests that the performance measurement at TQ organizations is a more participative process than at non-TQ organizations, where the process appears to be more top-down.

Performance management systems

Respondents were asked to identify which of a range of performance management techniques are used in their organization, including the use of performance appraisal, performance management, team management, the reward and recognition of teams and individuals, and linking the performance of processes and people.

TQ organizations were found to use performance measurement in performance appraisal, performance management, and linking people and processes more widely than non-TQ organizations. TQ and non-TQ organizations use performance measurement in team management and reward and recognition in similar proportions.

TQ organizations appear to use performance measurement more widely than non-TQ organizations in the management of individuals. This supports the findings from the case study organizations described in parts I and II[1,2], and also those found by Schneier *et al.*[19], who suggest that performance management is of vital importance in the management of TQ organizations.

Break-point performance assessment

Respondents were asked to identify which of a range of break-point assessment techniques were used by their organization. Techniques were separated into internal (involving resources and data within the organization) and external (sources or data from outside the organization).

Data analysis showed that TQ organizations generally use the range of internal break-point assessment techniques more widely than non-TQ organizations, particularly quality costing, self-assessment, business process re-engineering, and process capability analysis. Activity-based costing is used in similar proportions by TQ and non-TQ organizations.

As for internal techniques, TQ organizations generally use all external break-point assessment techniques (competitor/market analysis, benchmarking, customer surveys and quality function deployment) more widely than non-TQ organizations.

Effectiveness of performance measurement

Respondents were asked to rate the effectiveness of performance measurement within their organization. Data analysis showed that TQ organizations rated performance measurement to be more effective than non-TQ organizations at the 5 per cent level of significance.

Summary

The model developed was based on best practice performance measurement identified at the case study organizations. The survey was designed to examine some elements of the model. Despite the lack of statistical significance owing to the small number of non-TQ organizations responding, some general conclusions can be drawn about performance measurement at TQ and non-TQ organizations[6]:

- TQ organizations use a wider range of formal strategic management techniques than non-TQ organizations. TQ organizations appear to use non-financial performance measurement much more widely than non-TQ organizations at the organizational level.
- TQ organizations manage processes more widely than non-TQ organizations.
- TQ organizations use a wider range of performance measures at the process level than non-TQ organizations.
- TQ organizations use a wider range of break-point assessment techniques than non-TQ organizations.
- TQ organizations use non-financial performance measurement for more purposes than non-TQ organizations. As such, they generally use "fact-based" management more widely than non-TQ organizations.
- Performance measurement in TQ organizations appears to be more effective than in non-TQ organizations.

It is possible for non-TQ organizations to develop effective performance measurement systems. However, the nature of performance measurement systems at TQ organizations is different from non-TQ organizations. Also, it appears that TQ organizations are more likely to develop a truly effective performance measurement system than non-TQ organizations. In particular, the need to integrate strategic objectives, customer requirements, process capability and individual involvement is seen as vital for effective TQ-based performance measurement systems. The model developed allows for the introduction of an integrated performance measurement system, whereby individuals at all levels of the organization and all measurements are focused on the continuous improvement of processes towards increased customer satisfaction.

No attempt has been made to identify a prescriptive list of measures at each level of the performance measurement system model. Flexibility in the face of changing competition and customer requirements is vital if performance measurement is to remain supportive, and not become an inhibitor to organizational change[20]. The model therefore incorporates regular reviews of the measurement process at all levels, in order to ensure measurement is modified in line with changes in the competitive environment.

References

1. Sinclair, D. and Zairi, M., "Effective process management through performance measurement: part I – applications of total quality-based performance measurement", *Business Process Re-engineering & Management Journal,* Vol. 1 No. 1, 1995, pp. 75-88.
2. Sinclair, D. and Zairi, M., "Effective process management through performance measurement: part II – benchmarking total quality-based performance meaurement for best practice", *Business Process Re-engineering & Management Journal,* Vol. 1 No. 2, 1995, pp. 58-72.
3. Evangelidis, K., "Performance measured performance gained", *The Treasurer,* February 1992, pp. 45-7.
4. Pritchard, R.D., Roth, P.L., Jones, S.D. and Roth, P.G., "Implementing feedback systems to enhance productivity: a practical guide", *National Productivity Review,* Winter 1991, pp. 57-67.
5. Globerson, S., *Performance Criteria and Incentive Systems*, Elsevier Science Publications, Amsterdam, 1985.
6. Sinclair, D.A.C., "Total quality-based performance measurement systems: an empirical study of best practice", unpublished PhD thesis, Bradford University, 1994.
7. Rockart, J.F., "Chief executives define their own data needs", *Harvard Business Review*, March-April 1979, pp. 81-93.
8. Hronec, S.M., *Vital Signs: Using Quality, Time, and Cost Performance Measurements to Chart Your Company's Future,* Amacom, New York, NY, 1993.
9. Daniels, A.C. and Rosen, T.A., *Performance Management: Improving Quality and Productivity through Positive Reinforcement,* Performance Management Publications, Tucker, GA, 1988.
10. De Vries, D.L., *Performance Appraisal on the Line,* Centre for Creative Leadership, Greensborough, NC, 1986.
11. Walsh, F.J., "Current practices in measuring quality", *The Conference Board Research Bulletin No. 234,* The Conference Board, New York, NY, 1989.
12. Kaplan, R.S. and Norton, D.P., "The balanced scorecard – measures that drive performance", *Harvard Business Review*, January-February 1992, pp. 71-9.
13. Euske, K.J., Lebas, M.J. and McNair, C.J., "Performance management in an international setting", *Proceedings of the 16th Annual Congress of the European Accounting Association,* Turku, Finland, 28-30 April 1993.
14. Zairi, M., *Measuring Performance for Business Results,* Chapman & Hall, London, 1994.
15. Oakland, J.S., *Total Quality Management,* 2nd ed., London, Heinemann, 1993.
16. Hall, R.W., Johnson, H.T. and Turneym, P.B.B., *Measuring Up: Charting Pathways to Manufacturing Excellence*, Business One Irwin, Homewood, IL, 1991.
17. Hutt, G., "Incorporating quality objectives into performance appraisal systems", *TQM Magazine*, Vol. 6 No. 1, 1994, p. 8-12.
18 Longenecker, C.O., Scazzero, J.A. and Stansfield, T.T., "Quality improvement through team goal setting, feedback, and problem solving", *International Journal of Quality & Reliability Management,* Vol. 11 No. 4, 1994, pp. 45-52.
19. Schneier, C.E., Shaw, D.G. and Beatty, R.W., "Performance measurement and management: a tool for strategy execution", *Human Resource Management,* Vol. 30 No. 3, Autumn 1991, pp. 279-301.
20. Brignall, S., "Performance measurement systems as change agents: a case for further research", *Warwick Business School Research Papers No. 72*, Warwick Business School Research Bureau, 1993.